

PROJECT TITLE

by

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I declare that this dissertation is my own work and that the work of others is acknowledged and indicated by explicit references.

Bhavya Singh  
May 2026

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# Abstract

Write a summary of the work presented in your dissertation. Introduce the topic and highlight your main contributions and results. The abstract should be comprehensible on its own, and should not contain any references. As far as possible, limit the use of jargon and abbreviations, to make the abstract readable by non-specialists in your area. Do not exceed 300 words.

# Acknowledgements

Write any personal words of thanks here. Typically, this space is used to thank your supervisor for their guidance, as well as anyone else who has supported the completion of this dissertation, for example by discussing results and their interpretation or reviewing write ups. It is also usual to acknowledge any financial support received in relation to this work.

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# Glossary

|       |                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|
| $A_f$ | The source message, being a sequence of $f$ source symbols $a_1a_2 \dots a_f$                                                       |
| $a_i$ | The $i^{th}$ symbol in the source message, where $a_i \in S_m$                                                                      |
| $B_g$ | The decoded message, being a sequence of $g$ source symbols $b_1b_2 \dots b_g$                                                      |
| $b_i$ | The $i^{th}$ symbol in the decoded message, where $b_i \in S_m$                                                                     |
| $C_h$ | The transmitted (compressed) message, being a sequence of $h$ Tunstall codewords<br>$c_1c_2 \dots c_h$                              |
| $c_i$ | The $i^{th}$ codeword in the transmitted message, where $c_i \in T_n$                                                               |
| $D_h$ | The received (compressed) message, which for a complete Tunstall code is a sequence<br>of $h$ Tunstall codewords $d_1d_2 \dots d_h$ |
| $d_i$ | The $i^{th}$ codeword in the received message, where $d_i \in T_n$ for a complete Tunstall<br>code                                  |

# Abbreviations

|      |                                  |
|------|----------------------------------|
| BER  | Bit Error Rate                   |
| BPSK | Binary Phase Shift Keying        |
| BSC  | Binary Symmetric Channel         |
| DCT  | Discrete Cosine Transform        |
| ECC  | Error Correcting Codes           |
| FEC  | Forward Error Correction         |
| JPEG | Joint Photographic Experts Group |
| MPEG | Moving Pictures Experts Group    |
| SER  | Symbol Error Rate                |
| SNR  | Signal to Noise Ratio            |



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